

Exploratory Data Analysis

Exploratory Data Analysis (EDA)

The objective of this exploratory analysis is to understand the characteristics of the PhysioNet 2019 ICU cohort, investigate temporal behaviour, quantify informative missingness, and identify patterns that motivate the proposed deep learning architecture.

```
C:\Users\papaa\AppData\Local\Temp\ipykernel_13956\1385312637.py:9:
UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or
database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not
tested. Please consider using SQLAlchemy.
    patients = pd.read_sql(
C:\Users\papaa\AppData\Local\Temp\ipykernel_13956\1385312637.py:14:
UserWarning: pandas only supports SQLAlchemy connectable (engine/connection) or
database string URI or sqlite3 DBAPI2 connection. Other DBAPI2 objects are not
tested. Please consider using SQLAlchemy.
    measurements = pd.read_sql(
```

2. Dataset Overview

This section establishes the basic shape and structure of the cohort: how many patients, how many hourly observations, and what variables are available before any cleaning or modeling.

```
((40336, 6), (1552210, 37))
```

```
40336
```

```
1552210
```

```
Index(['patient_id', 'age', 'gender', 'unit1', 'unit2', 'hosp_adm_time'],
      dtype='str')
```

```
Index(['patient_id', 'iculos', 'hr', 'o2sat', 'temp', 'sbp', 'map', 'dbp',
      'resp', 'etco2', 'baseexcess', 'hco3', 'fio2', 'ph', 'paco2', 'sao2',
      'ast', 'bun', 'alkalinephos', 'calcium', 'chloride', 'creatinine',
      'bilirubin_direct', 'glucose', 'lactate', 'magnesium', 'phosphate',
      'potassium', 'bilirubin_total', 'troponini', 'hct', 'hgb', 'ptt', 'wbc',
      'fibrinogen', 'platelets', 'sepsis_label'],
      dtype='str')
```

```
patient_id      int64
icuclos         int64
hr              float64
o2sat           float64
temp            float64
sbp             float64
map             float64
dbp             float64
resp            float64
etco2           float64
baseexcess      float64
hco3            float64
fio2            float64
ph              float64
paco2           float64
sao2            float64
ast             float64
bun             float64
alkalinephos    float64
calcium         float64
chloride        float64
creatinine      float64
bilirubin_direct float64
glucose         float64
lactate         float64
magnesium       float64
phosphate       float64
potassium       float64
bilirubin_total float64
troponini       float64
hct             float64
hgb             float64
ptt             float64
wbc             float64
fibrinogen      float64
platelets       float64
sepsis_label    int64
dtype: object
```

3. Missingness Analysis (Informative Missingness)

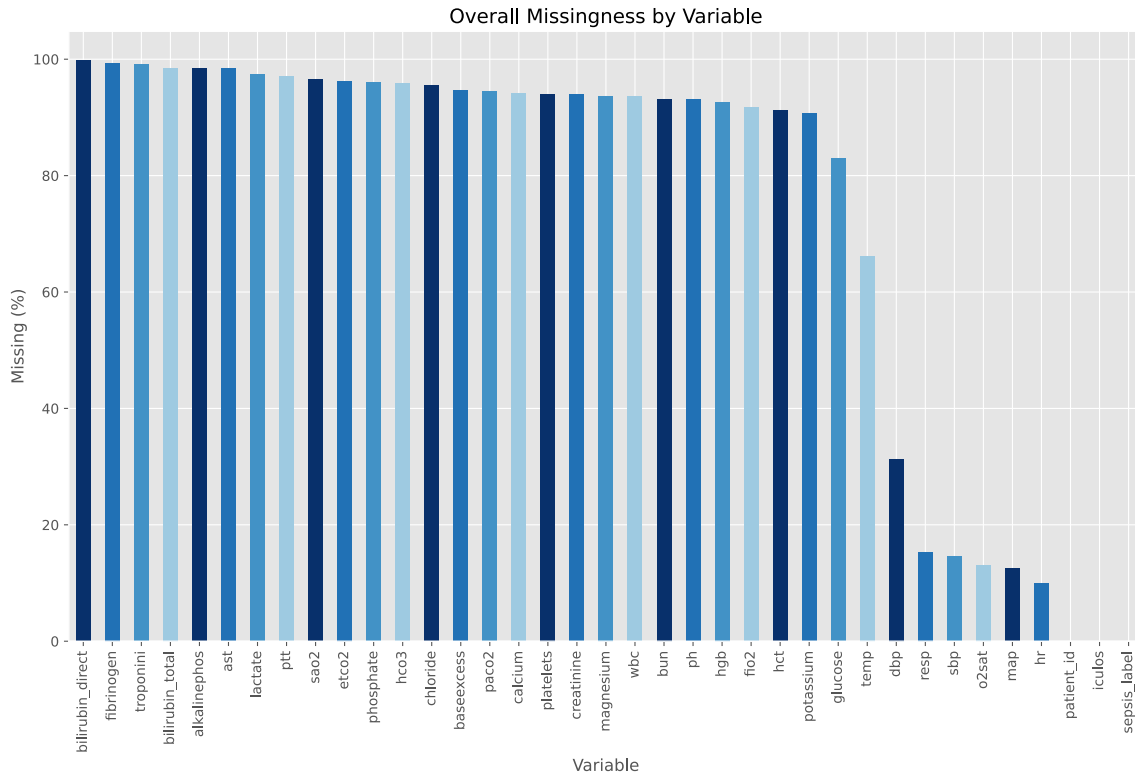
Traditional machine learning treats missing data as a nuisance to be imputed away.

Our hypothesis is different: in ICU environments, laboratory tests are ordered selectively by clinicians based on suspected patient status. Therefore, the absence of a measurement itself carries clinical information.

This section quantifies missingness across the PhysioNet cohort and motivates an informative missingness gating mechanism in the model architecture.

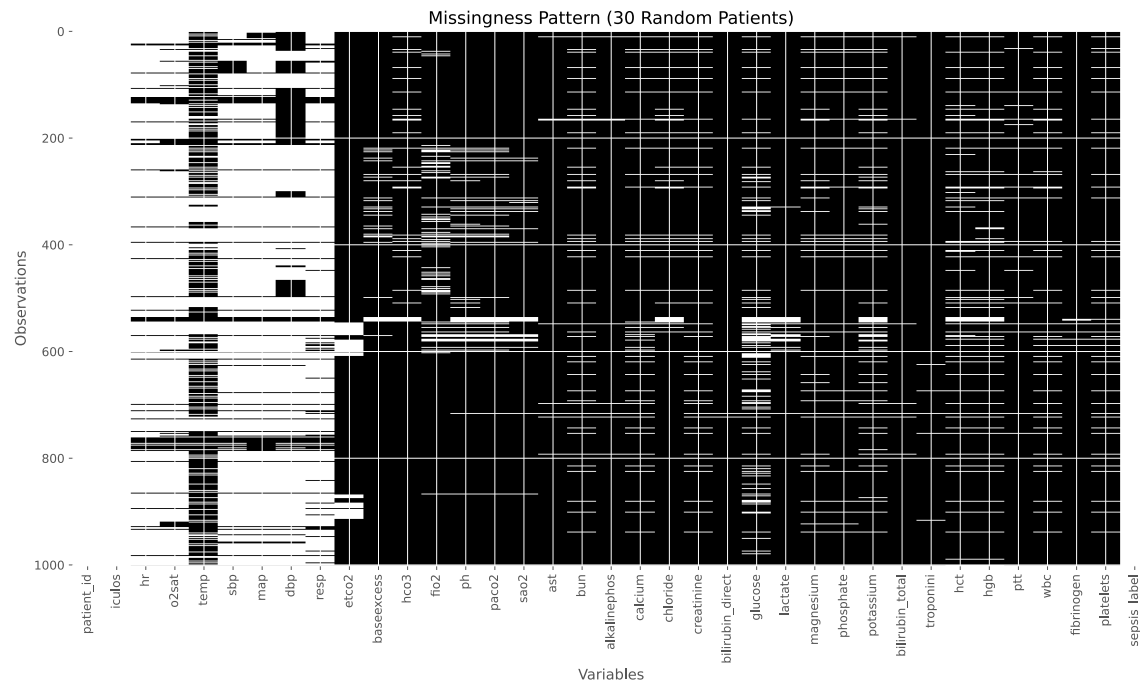
3.1 Overall Missingness

Bar chart of missingness across all clinical variables. This establishes the baseline case for a gating network that can learn from presence/absence patterns rather than treating missingness as pure noise.



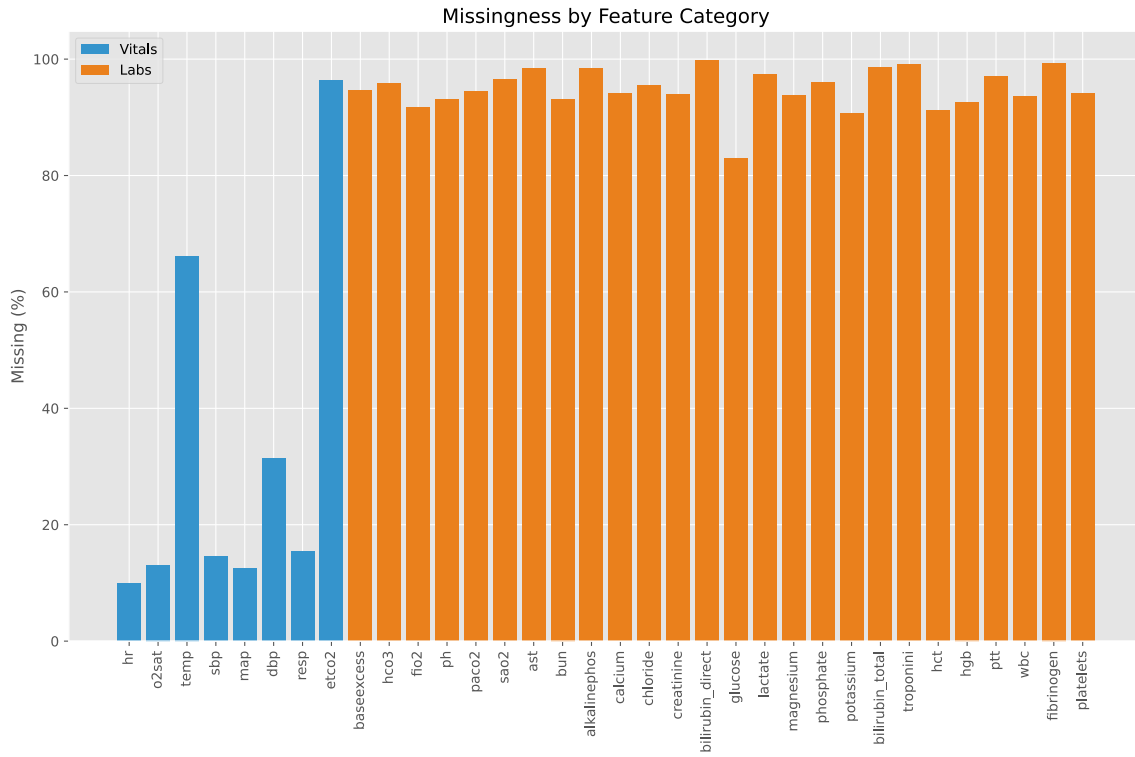
3.2 Missingness Heatmap

Missingness pattern for a random sample of patients, visualized as a binary matrix. This illustrates the irregular, non-random sampling intervals typical of ICU monitoring.



3.3 Missingness by Feature Category

Vitals (HR, MAP, SBP, Resp, Temp, O2Sat) are measured near-continuously, while labs (Lactate, Bilirubin, Creatinine, TroponinI, etc.) are ordered intermittently. This split shows that laboratory values disappear far more often than vital signs, which has direct implications for how each feature type should be handled.

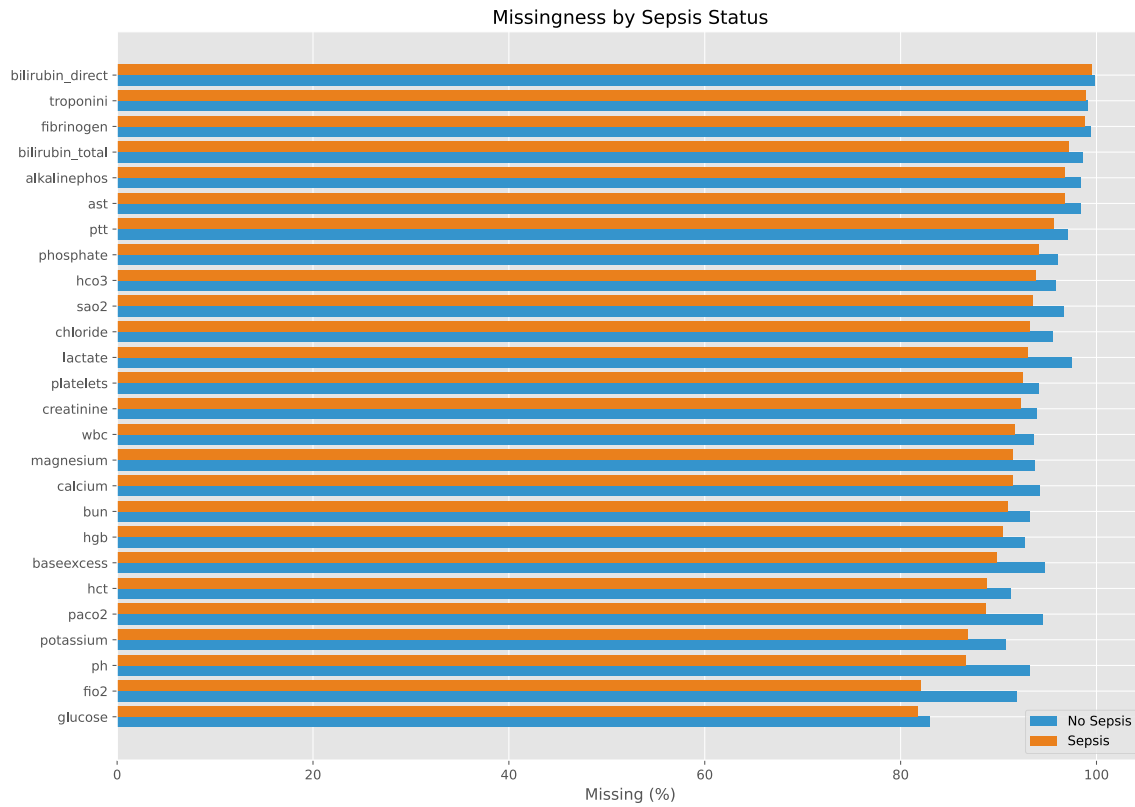


3.4 Missingness vs Sepsis

Are septic patients monitored more frequently than non-septic patients? If so, missingness itself becomes a predictive signal rather than noise to be discarded — the core justification for informative missingness gating.

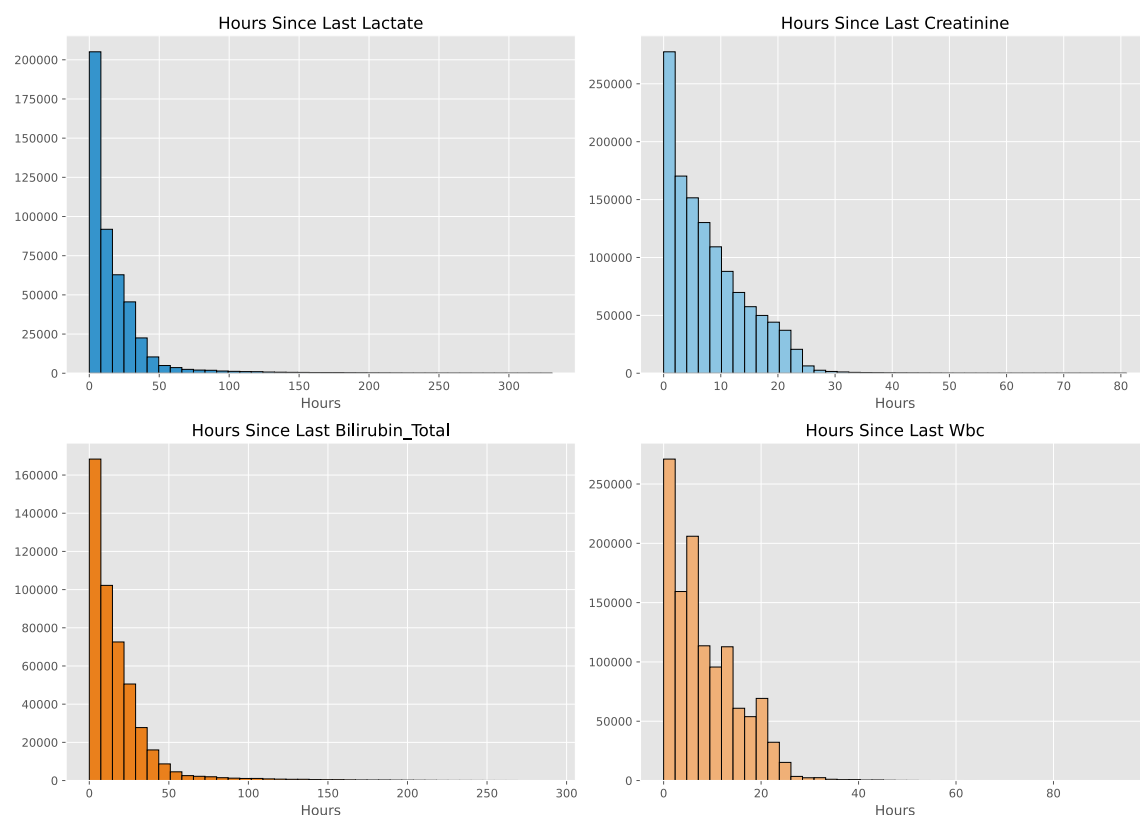
	No Sepsis (%)	Sepsis (%)
glucose	82.915041	81.763147
fio2	91.842453	82.017481
ph	93.187469	86.638487
potassium	90.760050	86.814013
paco2	94.545278	88.698238
hct	91.189364	88.777045
baseexcess	94.667105	89.769308
hgb	92.658503	90.381860
bun	93.174545	90.944261
calcium	94.166020	91.388451
magnesium	93.731065	91.424273

	No Sepsis (%)	Sepsis (%)
wbc	93.628395	91.671443
creatinine	93.934635	92.255337
platelets	94.088411	92.481014
lactate	97.409620	92.975355
chloride	95.502574	93.150881
sao2	96.604986	93.512681
hco3	95.847127	93.817166
phosphate	96.020715	94.107322
ptt	97.082256	95.615418
ast	98.407197	96.761714
alkalinephos	98.422811	96.776042
bilirubin_total	98.534666	97.116349
fibrinogen	99.349732	98.817882
troponini	99.051299	98.853704
bilirubin_direct	99.812372	99.534317



3.5 Time Since Last Measurement

For key deterioration markers (Lactate, Creatinine, Bilirubin, WBC), we compute hours elapsed since the last recorded value. This directly motivates a gating mechanism that weighs “staleness” of information, not just its presence or absence.

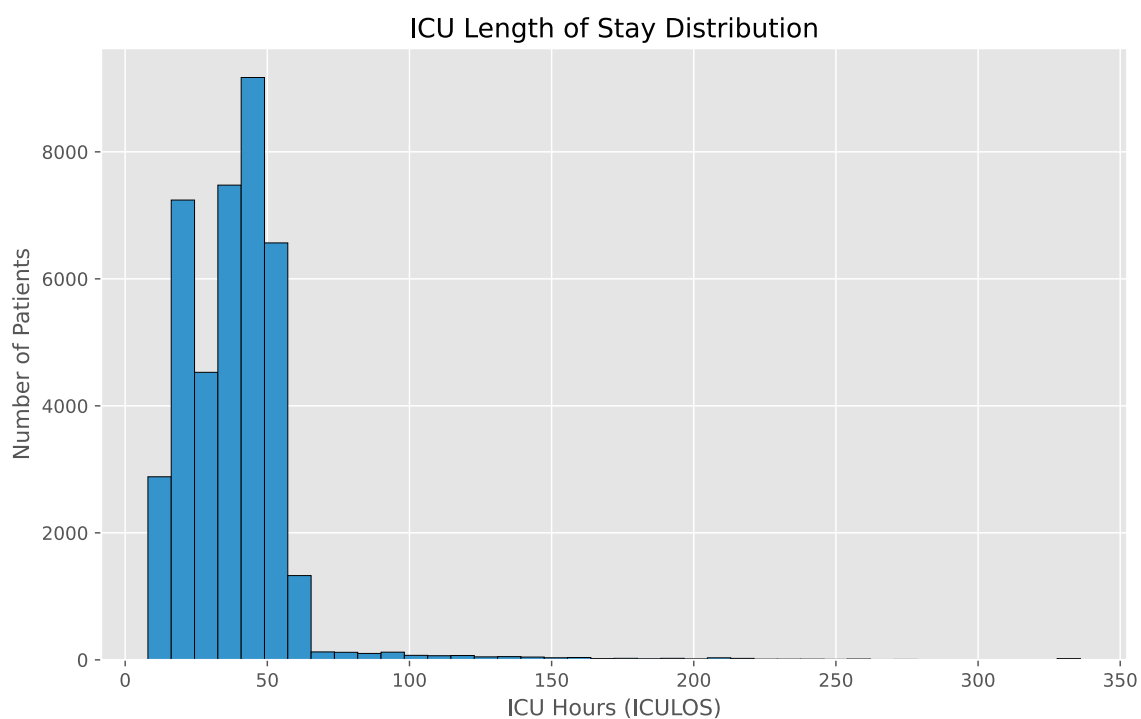


4. Temporal Dynamics

Having established how missingness behaves, we now turn from absence to the physiology itself: how vital signs evolve over an ICU stay, and whether their trajectories differ between patients who develop sepsis and those who do not.

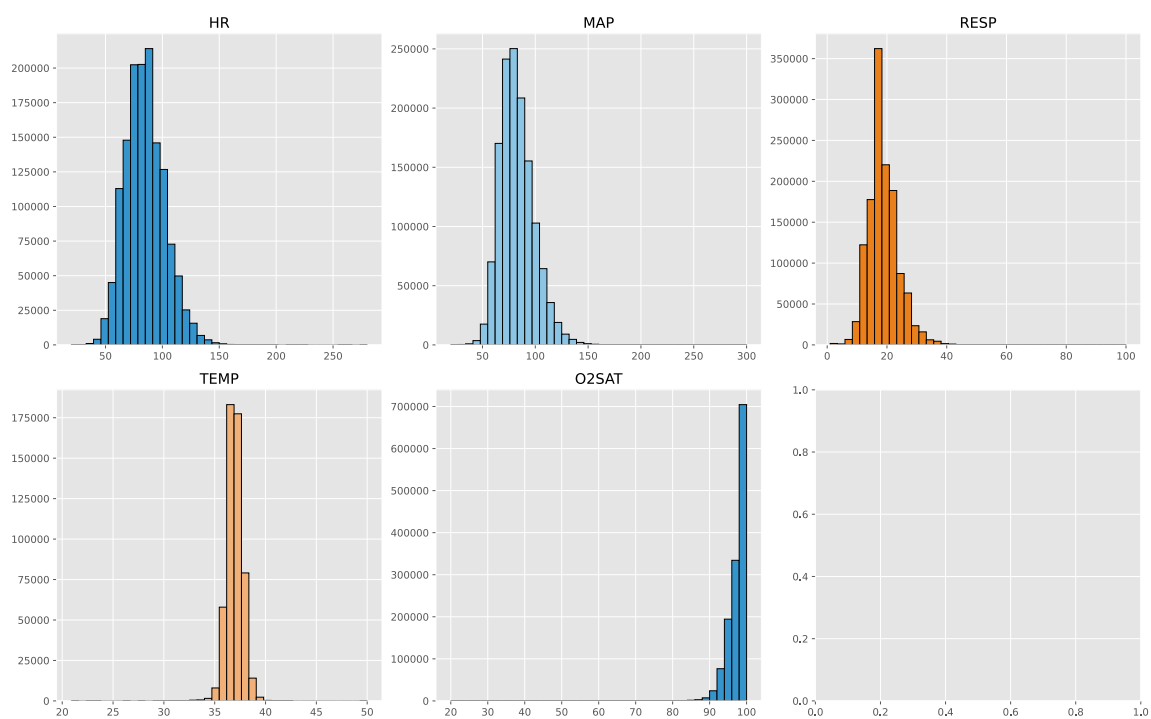
4.1 ICU Length of Stay Distribution

Histogram of ICULOS (ICU length of stay in hours) across the cohort. This informs the choice of sliding window size for the temporal model.



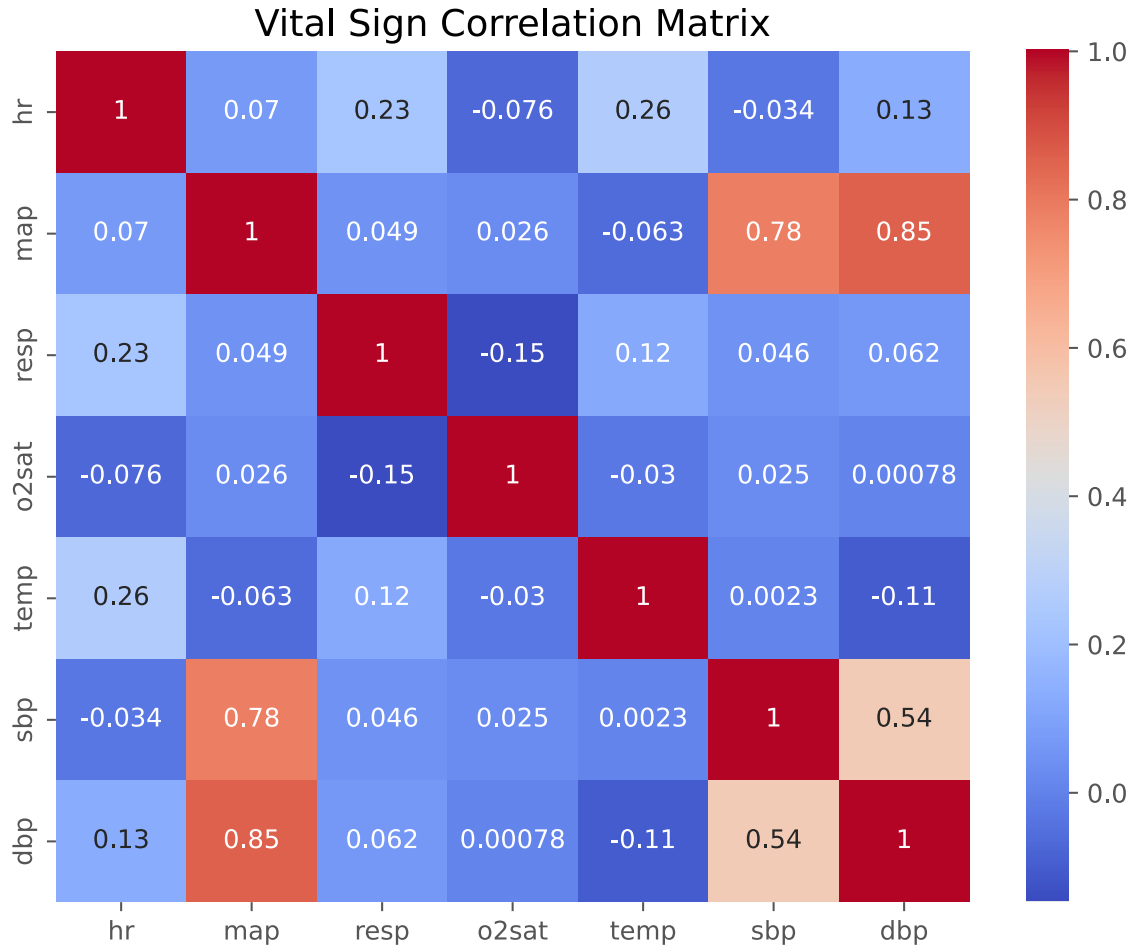
4.2 Vital Sign Distributions

Marginal distributions of HR, MAP, Resp, Temp, and O2Sat across all observations, establishing normal ranges and identifying outliers or skew before any modeling.



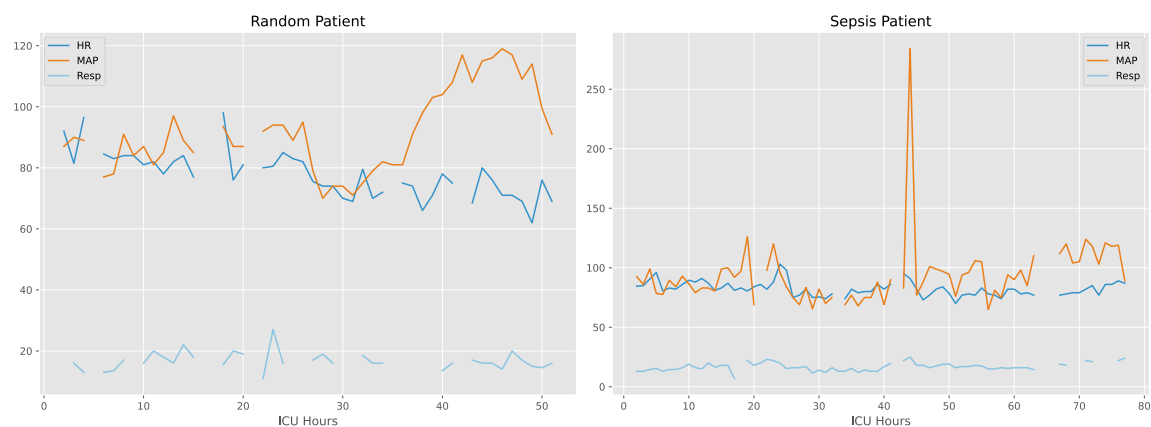
4.3 Correlation Matrix

Pairwise correlation between vital signs, checking for redundancy and physiologically expected relationships (e.g. HR and Resp).



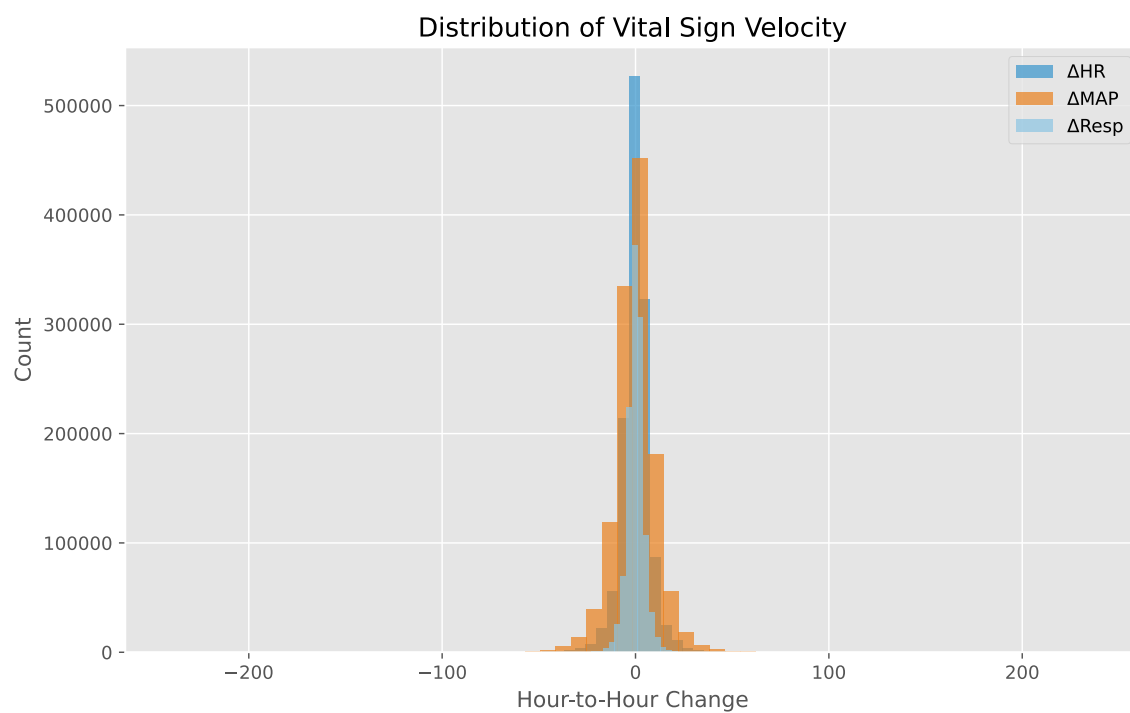
4.4 Patient Trajectories

Hour-by-hour HR, MAP, and Resp trajectories for a random patient versus a sepsis-positive patient, offering a first visual comparison of stability versus deterioration.



4.5 Velocity (Rate of Change)

Rather than raw vital sign values, we examine ΔHR , ΔMAP , and ΔResp — the hour-to-hour rate of change. Sepsis is a trajectory of deterioration, not a single abnormal reading, which is the central justification for a temporal convolutional (TCN) architecture over static feature models.



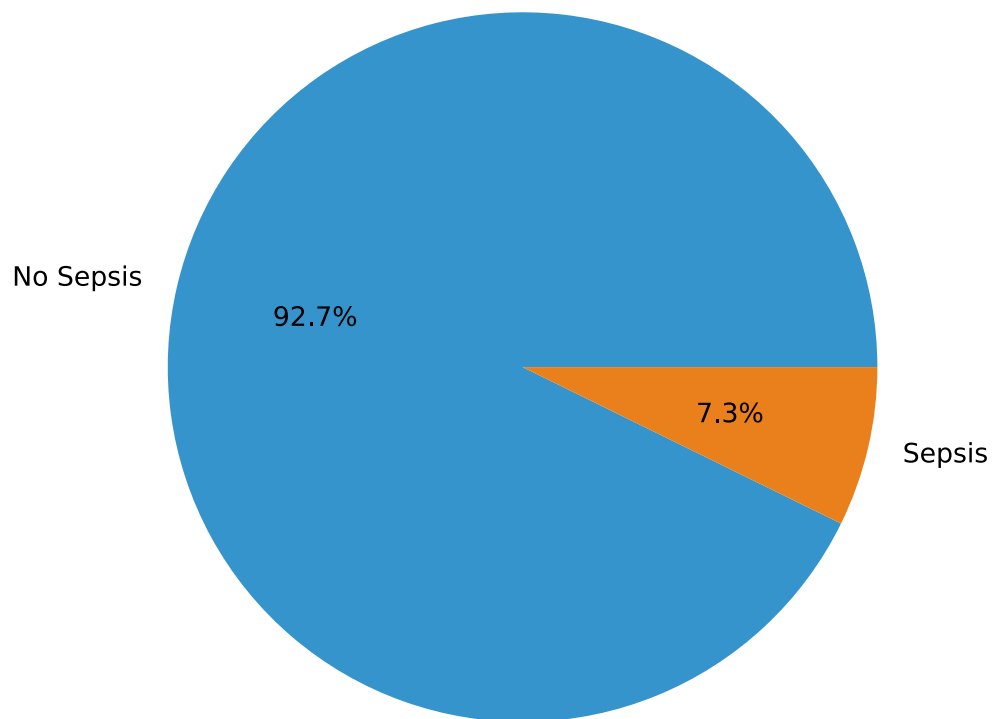
5. Sepsis Analysis

A dedicated section examining the target variable itself: how sepsis is distributed across patients, across hours, and over time relative to onset.

5.1 Patient-Level Class Balance

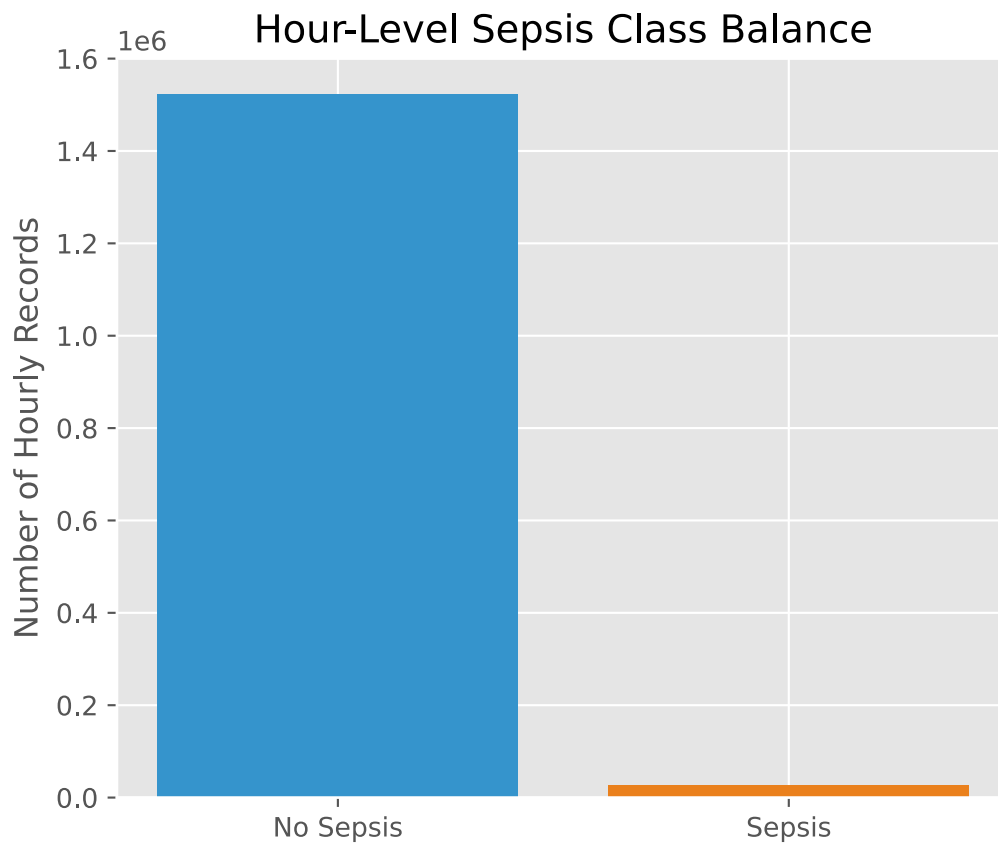
What proportion of patients in the cohort become septic at any point during their ICU stay?

Patient-Level Sepsis Class Balance



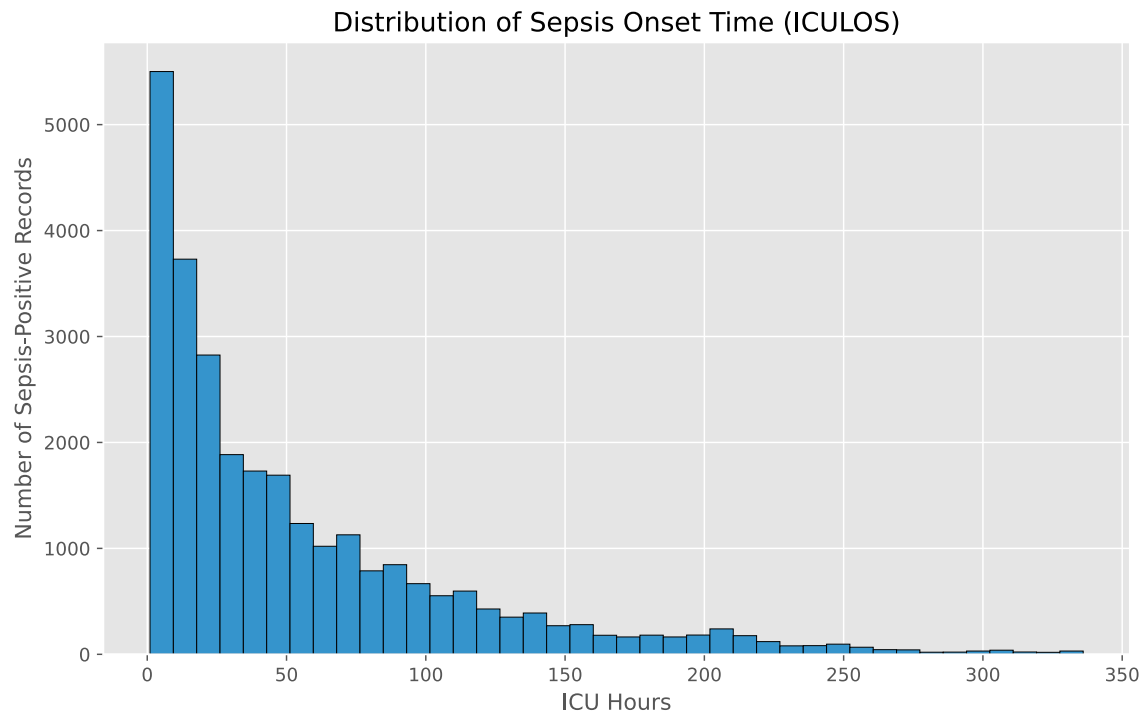
5.2 Hour-Level Class Balance

What proportion of individual hourly observations are labeled sepsis-positive? This is typically far more imbalanced than the patient-level split, and is the class balance that matters for model training.



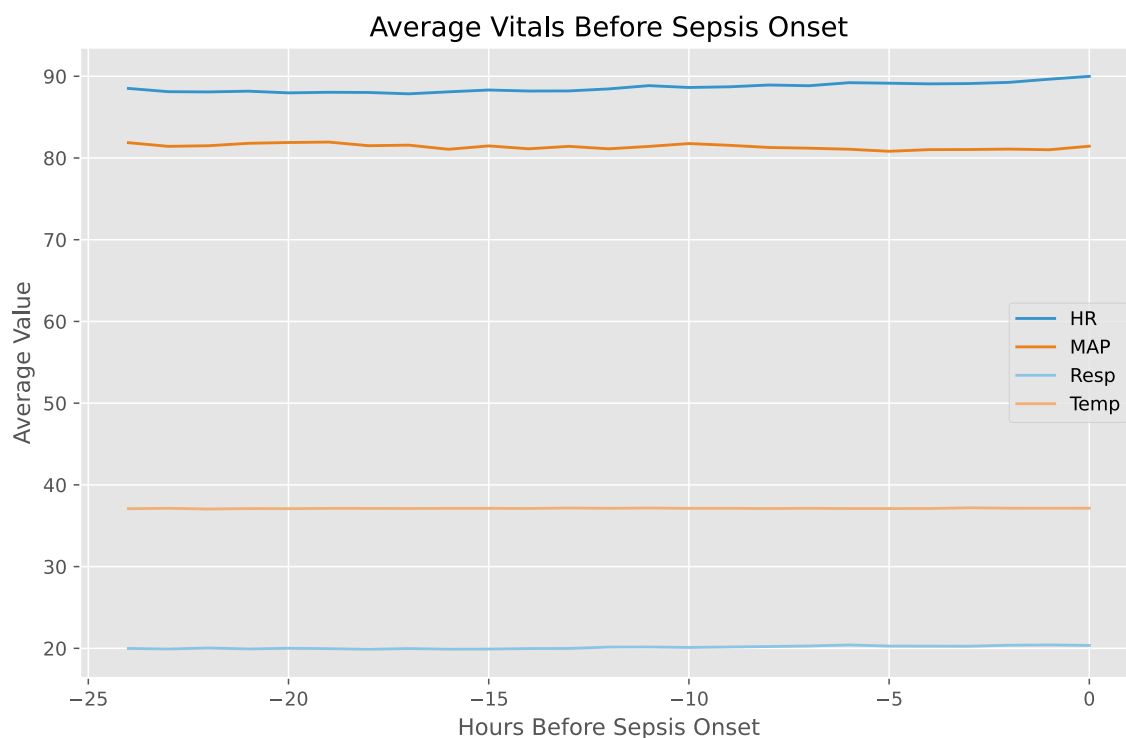
5.3 Sepsis Onset Distribution

Distribution of ICULOS at the hour sepsis is first flagged (SepsisLabel == 1). This shows when in a patient's stay onset tends to occur and supports the forecasting horizon design.



5.4 Hours Before Sepsis Onset

Average HR, MAP, Resp, and Temp in the hours leading up to sepsis onset, aggregated across septic patients. This traces the pre-onset deterioration pattern directly, supporting the case for trajectory-based embeddings over single-timepoint features.



6. Pre-Sepsis Physiological Drift

Section 5 showed **when** sepsis tends to occur. This section asks a different question: **how** a patient’s physiology behaves in the hours leading up to that onset.

If deterioration is gradual rather than a single abnormal reading, a model needs to see the *trajectory* of vitals, not just their most recent value. This is the empirical justification for a **Contrastive Trajectory Encoder (TCN)**: the architecture should learn to represent the shape of a patient’s recent physiological history, not a static snapshot.

6.1 Identify Sepsis Onset

For each patient, we locate the first ICU hour where `sepsis_label == 1`. This gives one onset hour per septic patient, which anchors all the “hours before onset” alignment that follows.

	patient_id	onset_hour
0	9	249
1	11	26
2	15	6
3	18	126
4	22	14

6.2 Build Pre-Onset Windows

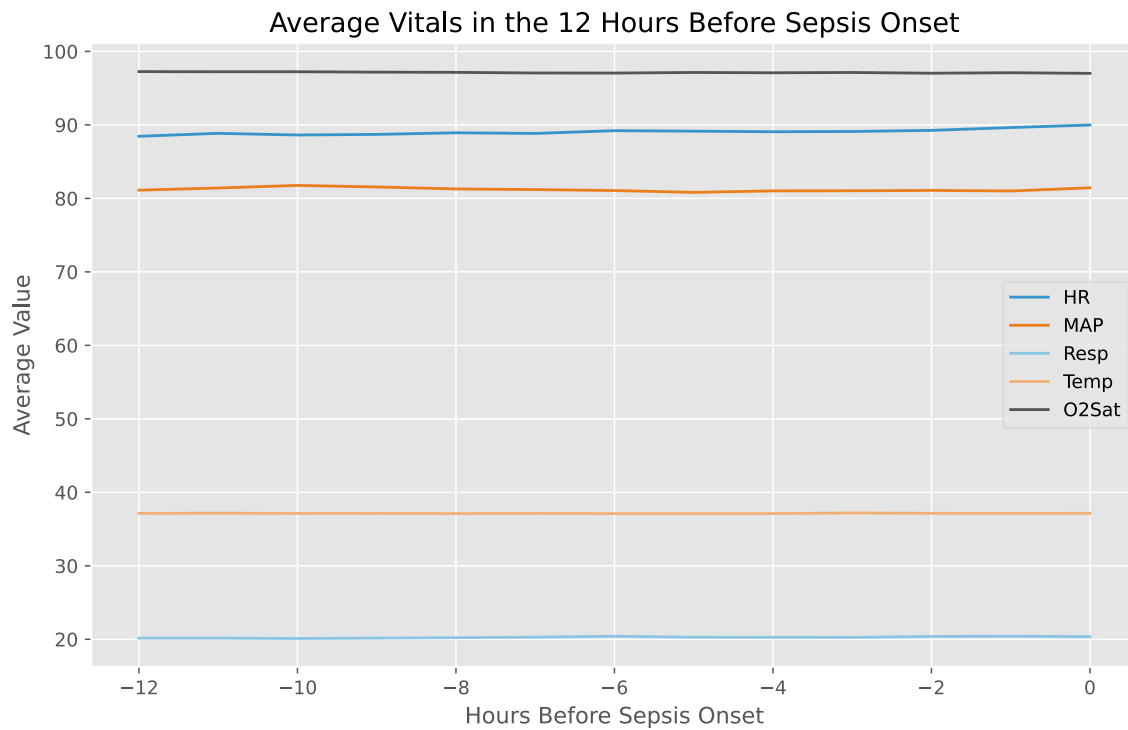
We align every septic patient's record to their own onset hour (`hours_before_onset = iculos - onset_hour`) and keep only the 12 hours immediately preceding onset. This is a shorter window than the 24-hour view in Section 5.4, focused specifically on the near-term deterioration horizon the model will need to forecast.

```
(29256, 38)
```

6.3 Average Pre-Onset Trajectories

For each hour in the 12-hour pre-onset window, we average HR, MAP, Resp, Temp, and O2Sat across all septic patients. A consistent drift in these averages — rather than a flat line followed by a jump — supports modeling physiology as a continuous trajectory rather than a set of independent, static features.

	hr	map	resp	temp	o2sat
hours_before_onset					
-12	88.453984	81.122319	20.153279	37.134007	97.254281
-11	88.851720	81.414734	20.162442	37.166436	97.233018
-10	88.626338	81.764504	20.106775	37.128809	97.236483
-9	88.708050	81.554728	20.165223	37.132003	97.182303
-8	88.929834	81.285804	20.219478	37.106579	97.150186
-7	88.841662	81.199601	20.288524	37.133011	97.066649
-6	89.213580	81.069839	20.410977	37.104033	97.055976
-5	89.148962	80.820414	20.277219	37.104352	97.136905
-4	89.069789	81.022924	20.266917	37.109149	97.094790
-3	89.105360	81.035058	20.257977	37.194623	97.134300
-2	89.255605	81.088863	20.381724	37.147024	97.036468
-1	89.641489	81.008982	20.416473	37.141721	97.101521
0	89.986298	81.440468	20.354784	37.148301	97.015767

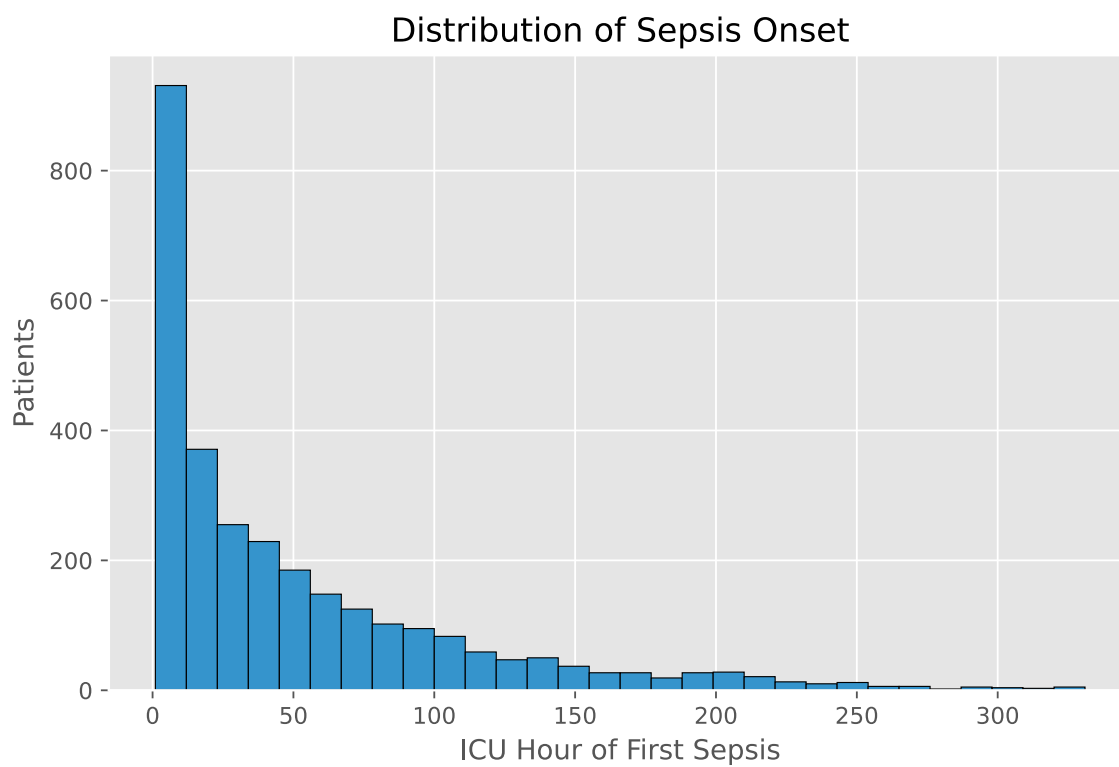


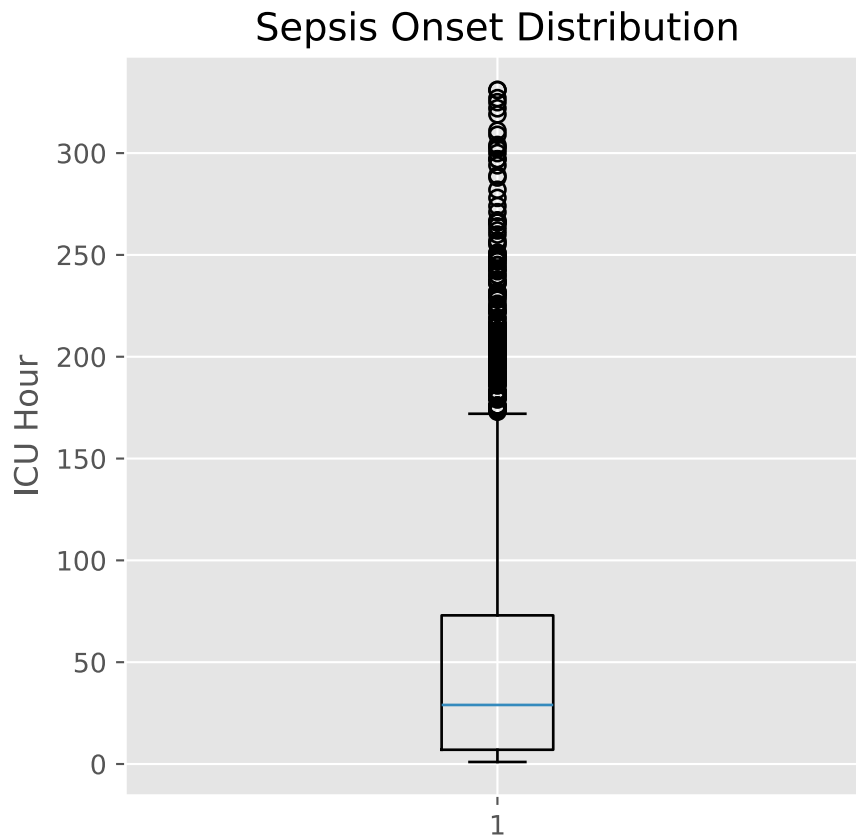
6.4 Sepsis Onset Timeline

Objective: confirm when, in ICU-hours, patients typically become septic.

Why it matters: the onset distribution directly supports the choice of a **sliding-window** input and a **12-hour forecasting horizon** — if onset were concentrated very early or very late in a stay, the windowing strategy would need to change accordingly.

```
count    2932.000000
mean      50.896317
std       59.363675
min        1.000000
25%        7.000000
50%       29.000000
75%       73.000000
max      331.000000
Name: onset_hour, dtype: float64
```





6.5 EDA-to-Architecture Mapping

This table summarizes how each empirical finding from the EDA translates into a specific design decision in the proposed deep learning architecture.

	EDA Finding	Architectural Decision
0	Extensive laboratory missingness	Missingness Gating Network
1	Missingness differs by clinical status	Informative Missingness Features
2	Vitals evolve gradually	Contrastive Trajectory Encoder (TCN)
3	Long ICU stays	Sliding Windows
4	Strong class imbalance	Weighted Binary Cross Entropy
5	Variable sepsis onset timing	12-hour Multi-Horizon Prediction